

GridWise Energy Optimization Challenge

Concise Summary of the BUP CSE Fest 2026 Preliminary Problem Statement

1. Challenge at a Glance

Build a public HTTP API that receives a 24-hour campus energy scenario and 1–3 natural-language operator notes. The service must use an LLM to interpret the notes, convert them into supported structured directives, validate them deterministically, apply them to an energy optimizer, and return a valid low-cost 24-hour schedule.

Item	Requirement
Round	Online preliminary; 7:00 PM–11:00 PM (4 hours)
Challenge type	LLM-assisted energy scheduling and optimization
Required API	Public HTTP API
Endpoints	GET /health and POST /optimize-energy
Planning horizon	24 hourly intervals (hours 0–23)
Operator notes	1–3 natural-language notes per scenario
Output	Structured JSON containing interpretation + schedule

2. Core System Flow

- **Input:** 24 hours of demand, solar availability, grid tariff, battery parameters, and operator notes.
- **LLM interpretation:** Convert every note into exactly one supported directive interpretation.
- **Validation:** Treat LLM output as untrusted data; deterministic guardrails check types, hours, values, and structure.
- **Optimization:** Apply all valid directives and find a schedule that minimizes grid electricity cost.
- **Replay/verification:** Recalculate the schedule hour-by-hour to ensure every rule and directive was actually followed.
- **Output:** Return both the machine-checkable interpretations and the complete 24-hour plan.

3. Supported Operator Directives

Directive	Meaning	Required adjustment
solar_reduction	Reduce usable solar in specified hours.	hours + factor (0–1)
minimum_battery_reserve	Keep battery energy at/above a specified level.	hours + minimum_energy_kwh
no_charge_window	Charging is forbidden in specified hours.	hours
no_discharge_window	Discharging is forbidden in specified hours.	hours
max_grid_window	Grid import cannot exceed a specified amount.	hours + max_grid_kwh
no_op	Note does not affect the current schedule.	null

4. Important Interpretation Rules

- Every operator note must produce exactly one interpretation entry, returned in note_index order.
- Only the six directive types above are allowed.
- Relevant notes must be applied to the optimization. Correct interpretation without application is still incorrect.
- Irrelevant notes must be **applies=false**, **directive_type=no_op**, and **structured_adjustment=null**.

- For all non-no_op directives, **applies=true**.
- Time windows use whole hours with an inclusive start and exclusive end. Example: 1 PM–3 PM → [13,14].
- Hours must be unique integers from 0–23 and sorted ascending.
- The LLM must not invent demand, solar, tariff, battery limits, or unsupported directive types.
- For solar_reduction, factor is the fraction remaining. An 80% reduction means factor = 0.2.

5. Optimization Objective

Goal: minimize total grid electricity cost after all valid operator directives have been applied.

```
total_cost_bdt = Σ (grid_kwh[h] × tariff_bdt_per_kwh[h]), for h = 0...23
```

A cheaper schedule is not acceptable if it violates an energy rule, battery constraint, or operator directive.

6. How Each Directive Changes the Model

- **solar_reduction:** effective_solar[h] = original_solar[h] × factor.
- **minimum_battery_reserve:** battery energy after the specified hours must meet the higher applicable reserve.
- **no_charge_window:** battery charge = 0 during listed hours.
- **no_discharge_window:** battery discharge = 0 during listed hours.
- **max_grid_window:** grid_kwh[h] ≤ specified maximum.
- **no_op:** no change to the optimization model.

7. API Contract

GET /health → HTTP 200 with {"status":"ok"} when ready.

POST /optimize-energy → accepts one scenario JSON object and returns interpretation + optimization plan.

Expected error behavior: 400 for malformed/structurally invalid JSON; 422 may be used for semantically invalid requests; 500 for controlled internal errors. Do not expose secrets or raw stack traces.

8. Request Structure

Field	What it contains
scenario_id	Unique synthetic scenario ID
operator_notes	1–3 non-empty natural-language notes
hours	Exactly 24 entries for hours 0–23
hours[].demand_kwh	Campus demand for that hour
hours[].solar_kwh	Base solar available before adjustments
hours[].tariff_bdt_per_kwh	Grid electricity price
battery.capacity_kwh	Maximum battery storage
battery.initial_energy_kwh	Energy at start of hour 0
battery.minimum_energy_kwh	Base minimum reserve
battery.max_charge_kwh_per_hour	Maximum hourly charge
battery.max_discharge_kwh_per_hour	Maximum hourly discharge

9. Battery and Energy Rules

- **Battery state:** charge $\rightarrow E_{\text{after}} = E_{\text{before}} + \text{battery_kwh}$; discharge $\rightarrow E_{\text{after}} = E_{\text{before}} - \text{battery_kwh}$; idle $\rightarrow$ unchanged.
- **Battery bounds:** $\text{minimum_energy_kwh} \leq E_{\text{after}} \leq \text{capacity_kwh}$. Active reserve directives can raise the minimum.
- **Hourly limits:** charge/discharge amounts cannot exceed their respective hourly limits.
- **Solar:** $0 \leq \text{solar_used_kwh} \leq \text{effective_solar_kwh}$. Unused solar is curtailed; grid export is not part of the challenge.
- **Energy balance:** $\text{grid} + \text{solar_used} + \text{battery_discharge} = \text{demand} + \text{battery_charge}$.
- **End-of-day neutrality:** final battery energy must equal initial battery energy.

10. Response Structure

- **scenario_id:** must match the request.
- **directive_interpretation:** one machine-checkable entry per operator note.
- **hourly_plan:** exactly 24 entries for hours 0–23.
- **total_grid_kwh:** sum of hourly grid purchases.
- **total_cost_bdt:** recalculated total grid cost.
- **peak_grid_kwh:** maximum hourly grid purchase.
- **plan_summary:** short human-readable explanation.

11. Hourly Plan Fields

Field	Requirement
hour	Integer 0–23
grid_kwh	Non-negative grid energy purchased
solar_used_kwh	Used solar; cannot exceed effective solar
battery_action	Exactly one: charge, discharge, idle
battery_kwh	Non-negative action magnitude; must be 0 when idle
battery_energy_after_kwh	Battery energy immediately after the hour

12. Hidden Evaluation: What Actually Matters

- Correctly determine whether each note is applicable or no_op.
- Extract the correct directive type, hours, and numeric values, including paraphrased wording.
- Return one interpretation per note, in the correct order, with no missing or duplicate mappings.
- Correctly apply the extracted directives to the schedule.
- Produce exactly 24 unique hours and valid non-negative numeric values.
- Respect battery transitions, capacity, minimum reserve, and charge/discharge limits.
- Never use more solar than the effective available amount.
- Satisfy the energy-balance equation for every hour.
- End with battery energy equal to the initial energy.
- Return totals (grid, cost, peak) that match values recalculated from the hourly plan.

13. LLM Guardrail Design — Practical Takeaway

The document explicitly separates language understanding from mathematical scheduling: the LLM handles natural-language interpretation, while deterministic code validates the LLM output and the optimizer handles the energy mathematics. A robust implementation should therefore avoid allowing free-form LLM output to directly control the optimizer.

Recommended internal pipeline: Notes → LLM structured output → schema/semantic validation → directive application → optimization → deterministic replay → JSON response.

14. Hidden Language Variation

Hidden cases may paraphrase the same directive. For example, statements about PV production dropping to one-fifth, panel washing leaving roughly one-fifth output, or an 80% rooftop-solar reduction all represent the same solar_reduction rule. The system should therefore interpret meaning rather than rely on exact phrase matching.

15. Numeric and Feasibility Rules

- Judge calculations generally accept values within 0.01 kWh or 0.01 BDT unless the official judge package specifies a stricter tolerance.
- Valid organizer scoring scenarios are feasible and do not require contradictory hard directives.
- Malformed or unsupported LLM output must be handled safely; do not invent new directive types or crash.
- Equivalent valid optimal schedules do not need to match a single reference JSON byte-for-byte.

16. Key Implementation Checklist

- Expose GET /health.
- Expose POST /optimize-energy.
- Validate incoming scenario structure and exactly 24 hours.
- Use an LLM in the actual operator-note interpretation path.
- Force structured directive output.
- Validate every LLM-generated directive deterministically.
- Support all five actionable directives plus no_op.
- Apply directives before optimization.
- Build a schedule satisfying energy balance and battery constraints.
- Enforce end-of-day battery neutrality.
- Recalculate/replay the final schedule before responding.
- Return all required response fields and accurate totals.

17. Bottom Line

The challenge is not simply an optimization problem and not simply an LLM task. It is a pipeline combining **natural-language understanding + deterministic validation + constrained energy optimization**. The judge checks both whether the notes were interpreted correctly and whether those interpretations were actually enforced in a mathematically valid 24-hour schedule. The separate Participant Guide & Evaluation Rubric remains the source for deployment, repository, submission, performance, scoring, penalties, and tie-break rules; those details are not defined by this problem statement.